Canonical notation, identifiers, and glossary

Originally written 2026-07-30 11:57 PDT; expanded to a full reference 2026-07-30 20:20 PDT; revised 2026-08-06 to settle N as the total across randomization paths; revised 2026-10-09 to merge the symbols of the whitepapers `docs/34` to `docs/58` and the mathematics compendium (see "Revision 2026-10-09" below)

This file is the single source of truth for mathematical notation, code identifiers, and controlled vocabulary across the manuscripts under `analysis/report/`. New manuscripts should adopt it rather than re-deriving symbols, and any deviation should be stated explicitly in the deviating paper's Methods section.

`analysis/report/whitepaper-notation-audit.md` was intended to hold a cross-reference audit of this symbol table against each manuscript's actual usage, but as of 2026-08-20 that file contains only YAML front matter and no audit content (pub_review whitepaper, Major Issue 2.5). Until that audit is written, this file's consistency with the eleven manuscripts rests on the compendium-wide `tools/notation-lint.pl` mechanized check (Minor Issue 3.9) rather than on a written audit trail.

Symbols and identifiers listed here were extracted from the manuscript sources, from `R/`, from the drivers under `analysis/scripts/`, and from the archived `.rds` summaries.

Revision 2026-10-09

The notation of the whitepapers `docs/34` to `docs/58` and of `analysis/report/compendium-mathematics.tex` was reviewed in five passes (2026-10-09) and the proposed symbols were merged here. Seven user decisions of 2026-10-09 settle the collisions the passes found:

1. τ^2 is the between-participant variance of the drug effect (treatment heterogeneity, as in Senn, paper 04, the paper 04 working protocol and `docs/58`). The residual variance of the E9 contrast given the biomarker, previously written τ^2 and τ_p^2 in `docs/37` to `docs/39`, the compendium and paper 01, is now σ_Δ^2 and $\sigma_{\Delta,p}^2$.
2. η_t is the design expectancy (1 open label, 0.5 blinded, 0 at baseline; code `e`, regressor `De`). A measured placebo-belief covariate, if one is used, is $\tilde{\eta}_{it}$. Paper 06's η(phase) is the design expectancy, not a measured quantity (paper 06 says so explicitly), so the former reading of η as "measured belief, paper 06" was withdrawn.
3. ρ is the AR(1) correlation per week; the common correlation of a compound-symmetry construct is ρ_{cs} (the published process has $\rho_{cs} = 0.8$).
4. The linear-time coefficient is β_T, multiplying the calendar week w_t.

5. The modified Gompertz curve is written with the mathematical symbols m (ceiling), d (displacement) and r (rate); the code names are `maxr`, `disp`, `rate`.
6. The share of the prognostic slope absorbed by a shared biomarker main effect is f. w_t is reserved for the calendar week, so row-group counts are n_{on}, n_{off}, n_v.
7. The prognostic couplings are $c_{bm,TV}$, $c_{bm,PB}$, $c_{bm,BL}$ (c_{bl} is retired).

Adopted without a decision, by consensus of the passes: the normal law is $\mathcal{N}(\cdot,\cdot)$, never $N(\cdot,\cdot)$; φ and Φ are the standard normal density and distribution function; $\mathbb{E}$ is the expectation operator; the remaining entries marked "(2026-10-09)" in the tables below. Where two documents still disagreed, the canonical meaning here won, and otherwise the compendium's choice won.

Part 1. Mathematical symbols

Each table gives the symbol, its meaning, its units ("none" for a dimensionless quantity, "points" for outcome-scale points), the code identifier where one exists, and where it is used. "CM" is `analysis/report/compendium-mathematics.tex`; "docs/NN" is a whitepaper under `docs/`; "all" means the manuscripts and whitepapers generally. Entries marked (2026-10-09) were added or changed in the merge described above.

Indices and structure

Symbol	Meaning	Units	Code	Used in
i	participant index	none	`ptID`	all
t	visit (timepoint) index; weekly measurement occasions throughout	none	row order within `ptID`	all
j, ij	period or occasion index within participant	none		papers 04, 06

Symbol	Meaning	Units	Code	Used in
w_t, w_s	calendar week of visit t (Hybrid: 4, 8, 9, 10, 11, 12, 16, 20); reserved for calendar time, never a weight (2026-10-09)	weeks	column `t` in drivers, `t_wk` in stored data	docs/34 to docs/58, CM
d_t	gap between visits, $w_t - w_{t-1}$ (2026-10-09)	weeks		docs/35, docs/36, CM
N	**total** number of participants in the trial, across all randomization paths; never a per-path count. See Part 2, sample size and randomization paths	participants	`N` (caller-dependent, Part 2)	all
P	number of randomization paths in a design; the expected allocation to each is N/P	none	`length(td$trial$paths)`	

Symbol	Meaning	Units	Code	Used in
q, p	randomization-path index, $q = 1, \ldots, P$, where a document also uses p for a p-value or an exponent (2026-10-09). The subscript p on path-specific quantities fixed by the 2026-10-09 decisions and the compendium ($\sigma^2_{\Delta,p}$, $\tilde{\beta}_p$, the path share ϖ_p) is retained; a document uses one of the two letters for paths throughout	none	`path`	q: docs/40, docs/57; p: docs/37, docs/58, CM
ϖ_p	share of participants allocated to path p, n_p/N ($1/P$ under equal allocation) (2026-10-09; formerly q_p in `docs/37` and the compendium)	none		docs/37, docs/58, paper 04 protocol, CM

Symbol	Meaning	Units	Code	Used in
n_{on}, n_{off}, n_v	numbers of on-drug and off-drug post-baseline visits (rows) of a path, and of rows of visit kind v (2026-10-09)	rows		docs/34, docs/37 to docs/39, docs/49, docs/55, docs/56
n_c	number of response components (3; 2 in papers 13 and 14) (2026-10-09)	none		CM
n_{reps}, n_{sim}	Monte Carlo replicates per cell	replicates	`n_reps`	all
k	cycles per participant in design sweeps; also the Weibull shape parameter in the decay family	none		papers 05, 12; CM

Outcome and latent components

Symbol	Meaning	Units	Code	Used in
Y_{it}	observed symptom score; the modeled outcome	points	column `Sx`	all
BL_i	participant-specific baseline level	points		all

Symbol	Meaning	Units	Code	Used in
BR_{it}	biological (pharmacological) response component	points		all
PB_{it}	placebo-belief response component	points		all
TV_{it}	time-variant natural-history component	points		all
ε_{it}	observation-level residual	points		all
u_i	participant random intercept	points	`random = ~1` by `ptID`	all
σ_u^2, σ_ε^2	variances of the random intercept and of the residual of the analysis model (2026-10-09; not σ_b^2, σ_e^2)	points^2		docs/41, docs/56, docs/57, CM
η_t	design expectancy at visit t: 1 open label, 0.5 blinded, 0 at baseline; paper 06 writes η(phase) for the same object (2026-10-09; formerly e_t)	none	column `e`; regressor `De`	papers 01, 06; docs/36 to docs/58, CM

Symbol	Meaning	Units	Code	Used in
$\tilde{\eta}_{it}$	measured (participant-reported) placebo-belief covariate, if an analysis uses one; reserved (2026-10-09)	none		docs/43
σ_{TV}, σ_{PB}, σ_{BL}	standard deviations of the natural-course and placebo-belief components and of the baseline level; $\sigma_{PB,t} = \sigma_{PB}\,\eta_t$ (2026-10-09)	points	`blparam`, `respparam` SDs	docs/35 to docs/38, docs/46, docs/49, docs/51, CM

Treatment exposure and carryover

Symbol	Meaning	Units	Code	Used in
D_{it}	binary drug state: 1 on drug, 0 off drug	none	`ondrug`	all

Symbol	Meaning	Units	Code	Used in
$D_{bc,it}$	continuous exposure-decayed drug indicator: 1 on drug, $e^{-\lambda t_{sd}}$ off drug after exposure, 0 at off-drug visits before the first exposure; in an analysis it is built at the assumed half-life $t^a_{1/2}$ (2026-10-09: pre-exposure value stated)	none	column `Dbc`	all
t_{sd}	time since discontinuation; input to the carryover decay	weeks	`tsd`	all
t_{od}	cumulative time on drug; input to the Gompertz response	weeks	`tod`	all
t_{pb}	time-by-placebo-belief accumulation	weeks	`tpb`	all
$t_{1/2}$	carryover half-life of the data-generating process	weeks (see Units)	`carryover_t1half`	all
$t^a_{1/2}$	half-life assumed by the analysis when it builds $D_{bc,it}$ (2026-10-09)	weeks	`t_half_a`; suffixes `Dbc0.5`, `Dbc1`, `Dbc2`	docs/43, docs/45 to docs/47, docs/56 to docs/58, CM

Symbol	Meaning	Units	Code	Used in
λ	carryover decay rate, $\ln 2 / t_{1/2}$	per week	`lambda_cor`	all
λ_w	Weibull rate, $(\ln 2)^{1/k} / t_{1/2}$	per week		CM
$\phi(\cdot)$, ϕ_t	decay-family weight (exponential, linear, Weibull, power) and its value at visit t, the residual fraction $e^{-\lambda t_{sd}}$ off drug after exposure and 0 before (2026-10-09)	none	`carryover_decay()`; `exp(-lambda_cor * tsd)`	docs/34, docs/36, docs/37, CM
$G(t_{od}; m, d, r)$	modified Gompertz mean of a component (Part 3), $G(0) = 0$ (2026-10-09)	points	`modgompertz`	docs/40, docs/46, docs/49, docs/51, docs/53, docs/55, CM
$\mu_{BR,t}$, μ_{last}	mean drug response at visit t after carryover; its value at the last on-drug visit, the anchor of anchored carryover (2026-10-09)	points	`brmeans`	docs/36, docs/37, docs/53 to docs/55

Symbol	Meaning	Units	Code	Used in
s_f	scale factor on the carryover exponent in the 2024 upstream code (commit 8609f12), default 2 (2026-10-09)	none	`scalefactor`	docs/53 to docs/55

Decay families implemented on the data-generating side: exponential $e^{-\lambda t}$; linear $\max(0, 1 - t/(2t_{1/2}))$; Weibull $e^{-(\lambda_w t)^k}$; power $\max(0, (1 - t/(3t_{1/2}))^p)$.

Biomarker and moderation

Symbol	Meaning	Units	Code	Used in
B_i	participant-level pre-treatment biomarker value	biomarker units	`bm`	all
b_i	standardized biomarker, $(B_i - \bar{B})/s_B$	none	`bm_z`	all
B_i^c	biomarker centered at the trial's baseline sample mean, $B_i - \bar{B}$, not scaled (2026-10-09)	biomarker units	`bmc`	docs/47, docs/56, docs/57, CM

Symbol	Meaning	Units	Code	Used in
μ_{bm}, σ_{bm}	population mean and standard deviation of the biomarker (μ_{bm} 2026-10-09)	biomarker units	`blparam`	docs/36 to docs/38, docs/48, CM
σ_{BR}	standard deviation of the drug-response component	points	`respparam`	all
c_{bm}	moderation parameter of the **covariance-moderation** architecture: on-drug correlation between biomarker and BR. A correlation, bounded in $[-1, 1]$	none	`c.bm`	all
β_{bm}	moderation parameter of the **mean-moderation** architecture: dimensionless multiplier scaling the treatment effect by b_i	none	`c.bm` with `dgp_architecture = 'mean_moderation'`	all

Symbol	Meaning	Units	Code	Used in
$c_{bm,A}$, $c_{bm,B}$	independent mean-channel and covariance-channel weights of the dual-channel architecture	none	`c.bm_a`, `c.bm_b`	paper 11
g_t, $\mathbf{g}$	coupling profile: the fraction of c_{bm} present at visit t, $\mathrm{Cor}(B_i, BR_{it}) = c_{bm}\, g_t$; 1 on drug; off drug after exposure ϕ_t (graded) or $\mathbf{1}[\mu_{BR,t} \neq 0]$ (step); 0 before first exposure (graded); $\mathbf{g}$ the vector over visits (2026-10-09)	none	built in `buildSigma()`	papers 01, 11; docs/34 to docs/37, docs/40, docs/41, docs/44, docs/49, docs/51, docs/55, CM
$\tilde{g}$	the coupling profile padded with zeros at the TV and PB entries (2026-10-09)	none		docs/36, CM

Symbol	Meaning	Units	Code	Used in
$\bar{g}_{\text{on}}$, $\bar{g}_{\text{off}}$, $\mathcal{G}$	mean coupling over a path's on-drug and off-drug visits, and the coupling gap $\mathcal{G} = \bar{g}_{\text{on}} - \bar{g}_{\text{off}}$ (2026-10-09)	none		docs/40, docs/49, docs/55, CM
c^*_{bm}	coupling ceiling: the largest c_{bm} for which every path's joint correlation matrix is positive definite, $(\tilde{g}^\top M^{-1} \tilde{g})^{-1/2}$ (2026-10-09)	none		docs/34 to docs/37, docs/39, docs/44, docs/54, CM
$c_{bm,TV}$, $c_{bm,PB}$, $c_{bm,BL}$	prognostic couplings: correlation of the biomarker with every post-baseline TV element, every post-baseline PB element, and the baseline level (2026-10-09; c_{bl} retired)	none	`c_tv`, `c_pb`, `c_bl` (stress-test driver); `c.bm.pb` (package)	docs/35, docs/44, docs/46 to docs/49, docs/51, docs/56, docs/57, CM

Symbol	Meaning	Units	Code	Used in
s_{prog}, f	prognostic slope of Y on the biomarker that does not depend on drug, and the share of it absorbed by a shared biomarker main effect β_B (2026-10-09; not q)	points per biomarker unit; none		docs/48, docs/49, docs/51, docs/56, CM

Estimand, inference, and performance

Symbol	Meaning	Units	Code	Used in
$\beta_{bm:D}$	the target estimand: the biomarker-by-treatment interaction coefficient	points per biomarker unit	`bm:Dbc`	all
β_{bm}^{BR}, β_{bm}^{PB}, β_{bm}^{TV}	component-specific biomarker slopes (paper 06); β_{bm}^{TV} is also the stress-test DGP parameter for a mean moderation of TV (docs/44)	points per biomarker unit		paper 06, docs/44, CM

Symbol	Meaning	Units	Code	Used in
$\beta_{bm}^{\text{lumped}}$	the one-component (single-indicator) biomarker slope	points per biomarker unit		paper 06
β_D	treatment main effect	points	coefficient of `Dbc` or `D`	papers 04, 05; docs/56 to docs/58, CM
τ^2	between-participant variance of the drug effect: treatment heterogeneity (Senn); in the paper 04 protocol the variance of the participant deviation v_i of the drug-response ceiling (2026-10-09)	points^2	`tau_squared`, `tau`	paper 04, its protocol, docs/58, CM (main-effect track)
θ, $\hat{\theta}$	a generic interaction coefficient; used only where the argument is deliberately not N-of-1 specific	points per biomarker unit		paper 10, CM
θ_{true}	calibrated true value, $-c_{bm}\,\sigma_{BR}$	points per biomarker unit	`true_value`	paper 10

Symbol	Meaning	Units	Code	Used in
κ	ratio of mean model-based standard error to empirical standard deviation; $\kappa > 1$ means a conservative test	none		papers 08, 10
δ	standardized bias of the test statistic	none		paper 10
ν	denominator degrees of freedom	none		all
α	nominal significance level; 0.05 throughout	none		all
π, $\hat{\pi}$	power, $\Pr(p < \alpha)$, and an estimated rejection rate, with $\text{MCSE}(\hat{\pi}) = \sqrt{\hat{\pi}(1-\hat{\pi})/n_{\text{reps}}}$ ($\hat{\pi}$ 2026-10-09)	none	`power`, `mc_se_power`	all
γ	a $BR \times PB$ interaction coefficient, the subadditivity/superadditivity term	points		paper 06

Symbol	Meaning	Units	Code	Used in
ψ_g	latent-class gating slope: steepness of $\pi(B_i)$'s logistic dependence on the standardized biomarker (called `gs` in code and earlier prose)	none	`gs`	paper 03
Z_i	unobserved latent class label; also the responder label of a responder mixture (docs/51)	none		paper 03, docs/51

Analysis-model, covariance-construct and stress-test symbols

Added 2026-10-09. Coefficients of the analysis model (written in R as `Sx ~ bmc * (Dbc + De) + t`, with variants), of the paired-difference statistic E9, and generic conventions.

Covariance construct

Symbol	Meaning	Units	Code	Used in
ρ	AR(1) within-component correlation per week, Cor = $\rho^{\lvert w_t - w_s \rvert}$ (refined process 0.7; configuration C in `docs/36` 0.8)	per week	`rho`	all
ρ_{cs}	common within-component correlation of a compound-symmetry construct, the same at every lag (published process 0.8) (2026-10-09)	none	`rho` in the published code	docs/34 to docs/36, docs/44, docs/48, docs/51, docs/54, CM
ρ_t	AR(1) step correlation between visits $t-1$ and t, ρ^{d_t}	none		docs/35, docs/36, CM
c_1, $c_\times$	cross-component correlation at the same visit and at different visits (the latter times $\rho^{\lvert w_t - w_s \rvert}$ under the separable form)	none	`c.cf1t`, `c.cfct`	docs/34 to docs/37, docs/40, docs/44, docs/50, docs/51, docs/54, CM

Symbol	Meaning	Units	Code	Used in
K, A, M	$n_c \times n_c$ cross-component correlation, temporal correlation ($A_{ts} = \rho^{\lvert w_t - w_s \rvert}$ under AR(1)), and the response block, $M = K \otimes A$ when separable	none		docs/34 to docs/37, docs/39, docs/44, docs/51, CM

Analysis model, contrasts and conventions

Symbol	Meaning	Units	Code	Used in
β_0	intercept of the analysis model	points	`(Intercept)`	docs/56, docs/57
β_B	biomarker main effect (prognostic coefficient of the raw or centered biomarker); distinct from the mean-moderation multiplier β_{bm}	points per biomarker unit	coefficient of `bm` or `bmc`	docs/40, docs/41, docs/43, docs/45, docs/46, docs/48, docs/49, docs/51, docs/56, docs/57, CM
β_T	coefficient of linear time, $\beta_T\, w_t$ (not β_t)	points per week	coefficient of `t`	docs/40, docs/42, docs/45, docs/56, docs/57, CM

Symbol	Meaning	Units	Code	Used in
μ_t	visit effect, the free mean of visit t when time enters as a factor	points	`factor(t)`	docs/40, docs/42
β_{η}, $\beta_{bm:\eta}$	coefficients of the design expectancy and of the biomarker-by-expectancy term	points; points per biomarker unit	`De`, `bmc:De`	docs/40, docs/42, docs/49, docs/56, docs/57, CM
$\beta_{B,0}$	extra biomarker slope at baseline	points per biomarker unit	`bmc:bl0`	docs/49, CM
β_{BL}, $\beta_{BL,t}$	coefficient on baseline in the covariate coding, and the slope of visit t on baseline	none		docs/41, CM
ρ_a	working CAR(1) correlation per week of the analysis model, distinct from the data-generating ρ	per week	`corCAR1` parameter `Phi`	docs/56 to docs/58, CM
Δ_i, a_t	participant i's on-minus-off contrast $\sum_t a_t Y_{it}$ (E9) and its visit weights $1/n_{\text{on}}$, $-1/n_{\text{off}}$	points; none		papers 01, 02; docs/37 to docs/41, docs/43, docs/55, CM

Symbol	Meaning	Units	Code	Used in
σ^2_Δ, $\sigma^2_{\Delta,p}$	residual variance of Δ_i given the biomarker, overall and in path p (2026-10-09; formerly τ^2, τ^2_p)	points^2		paper 01; docs/37 to docs/39, CM
$r_{\Delta B}$	correlation between the contrast Δ_i and the biomarker (not $\rho_{\Delta B}$)	none		paper 01; docs/37 to docs/39, CM
Λ	noncentrality parameter of a t or F test	none		paper 01; docs/37, docs/58, CM
L_{it}	lagged just-off-drug indicator of specification E3, $D_{i,t-1}(1 - D_{it})$	none		docs/43, CM
ℓ_g	eigenvalue weights of the CR2 Satterthwaite degrees of freedom	none		docs/56, CM
$\mathcal{N}(\mu, \sigma^2)$	the normal law; never $N(\cdot)$, since N is the total sample size		`rnorm`	all
$\varphi(\cdot)$, $\Phi(\cdot)$	standard normal density and distribution function (ϕ is the decay family)	none	`dnorm`, `pnorm`	docs/51, docs/58, CM

Symbol	Meaning	Units	Code	Used in
$\mathbb{E}$	expectation operator (distinct from specification labels E1 to E9)			docs/51, CM

Reserved letters (2026-10-09). Besides the letters of the tables above, these may not be reused for another meaning in a document that also uses the canonical one: w (calendar week; use n for row counts and another letter for weights), f (absorbed share; a drug-response time course is $h(t)$, as in `docs/44`), τ (heterogeneity variance), η (design expectancy), ρ (AR(1) correlation; ρ_{cs}, ρ_t, ρ_a are its decorated relatives), ϕ (decay family) against φ (normal density), q (path index, or p as in the q, p row), g (coupling profile), Δ (E9 contrast), and m, d, r when they appear undecorated beside a Gompertz curve.

The five rules the symbols encode

1. D_{it} **is binary;** $D_{bc,it}$ **is continuous.** These are different objects, and the distinction is the subject of the carryover-specification manuscript. Never use one for the other.
2. c_{bm} **is a correlation;** β_{bm} **is a multiplier.** They are calibrated to a common numeric scale, because the mean-channel shift is applied as $\beta_{bm}\, b_i\, \sigma_{BR}$ on the standardized biomarker, so the reference value 0.45 denotes a matched moderation strength under either architecture. The symbols are nonetheless not interchangeable. A paper reporting both architectures at a matched value may use one shared label, but must say so.
3. **Mathematics in model definitions, code identifiers in code contexts.** Write $Y_{ij} = \beta_0 + \beta_D D_{bc,ij} + \ldots$ for the model; write `Sx ~ bm + t + Dbc + bm:Dbc` in `\texttt{}` when quoting the R formula. Never mix the two inside one expression.
4. N **is the total sample size.** N is the total number of participants across all randomization paths, and the expected allocation to each of P paths is N/P: with $P = 4$ and $N = 140$, each path receives about 35. Never use N for a per-path count. A paper may name a per-path count alongside the total, in words or as N/P, but must give N itself as the total. Because two of the drivers fix the per-path count instead, the reported N cannot be taken on trust: check the driver against the table in Part 2 before quoting a sample size.
5. **State the sign convention.** The three components are non-negative *reductions* in symptom severity, so an increase in any component lowers Y.

Treatment effects and interaction coefficients are consequently negative; moderation parameters are positive.

What the linter checks

`tools/notation-lint.pl` mechanizes the part of this file that can be checked from the manuscript sources. Run it over the manuscripts:

```
perl tools/notation-lint.pl analysis/report/*/report.Rmd
```

It skips fenced R chunks and `verbatim` blocks and strips `\texttt{}` and backtick spans (and, in `.tex` files, the compendium's `\code{}` macro), including a `\texttt{}` or `\code{}` span that continues across a line break, so it sees reader-facing prose and mathematics only (2026-10-09: `\code{}` and multi-line spans added). Four checks are implemented.

Check	Fires on	Rule
`bare-Dbc`	`Dbc` as a symbol outside a code span	1
`bare-Sx`	`Sx` as a symbol outside a code span	3
`legacy-spec-label`	`A1`, `A2`, `A3`	Part 3, specifications
`N-per-path`	a value of N directly qualified as per-path	4

`N-per-path` fires on `$N = 35$ per randomization path` but not on `$N = 140$ (70 per path)`, which is the compliant way to give both numbers. A day-denominated $t_{1/2}$ is exempted when the file carries a `**Units.**` paragraph (or, in LaTeX, a `\textbf{Units.}` paragraph), which is how rule 3's units requirement is enforced.

Rule 2 and rule 5 are not mechanized: whether a shared c_{bm} label is declared, and whether the sign convention is stated, both need a reader. One known false positive stands, a `bare-Sx` inside a backtick span that crosses a line break in paper 02.

Units

Carryover half-life $t_{1/2}$ is quoted in **weeks** in the interaction-focused manuscripts, on the canonical grid $\{0, 0.5, 1.0\}$, and in the main-effect manuscript (paper 04); it is quoted in **days** in the test-procedure manuscripts, on a pharmacokinetic scale with a 3-day baseline. Both conventions are retained because each paper's figures are drawn on its own scale. Every paper using days carries an explicit units paragraph giving the conversion (1 week = 7 days). Timepoints t are weekly in both conventions.

Revised 2026-10-09 (user decision): paper 04 is being rebuilt under the refined process and uses weeks, as the interaction manuscripts do (`analysis/scripts/treatment-main-effect/00-ademp-protocol.md`, "Units"). The day scale with a 3-day baseline belonged to the version 1 study of paper 04 (`00-ademp-protocol-v1.md`), which this paragraph previously covered.

Part 2. Code identifiers and stored data values

These are the names as they appear in `R/`, in the drivers, and in the archived `.rds` summaries.

Where a stored value is also a reporting label, the two are kept **identical**, so that no display-time mapping stands between the data and the page. A mapping layer is a standing hazard: it has to be applied at every display site, it fails silently when a rename reaches one layer and not the other, and it makes a stored table unreadable without the manuscript beside it. Where a stored value is a package API argument (`dgp_architecture = 'mvn'`), it stays as the API defines it and the manuscript maps it to a reporting name, which is the ordinary factor-level-to-label relationship.

Exported package functions

`buildtrialdesign`, `buildSigma`, `generateData`, `censordata`, `lme_analysis`, `generateSimulatedResults`, `validateParameterGrid`, `cumulative`, `modgompertz`, `trajectoryShape`, `shape_logistic`, `shape_hyperbolic_tangent`, `shape_piecewise_linear_breakpoint`, `characterize_carryover`, `analyze_trial_extended`, `print_carryover_summary`, `print_trial_summary`, `plotfactortrajectories`, `PlotModelingResults`, `reknitsimresults`.

Principal arguments

Identifier	Role
`modelparam`	list: sample size `N`, moderation `c.bm`. The meaning of `N` depends on the caller; see sample size and randomization paths below
`respparam`	Gompertz parameters `maxr`, `rate`, `disp` per component
`blparam`	biomarker and nuisance means and standard deviations

Identifier	Role
`trialdesign`	output of `buildtrialdesign`: `timepoints`, `timeptnames`, `expectancies`, `ondrug`
`dgp_architecture`	`'mvn'`, `'mean_moderation'`, or `'combined'`
`carryover_t1half`	carryover half-life
`lambda_cor`	decay rate; derived from the half-life when `NA`
`br_family`	response-curve family: `'gompertz'` and alternatives
`br_p2`, `br_p3`	shape parameters of the alternative families
`moderation_scaling`	`'constant'` or trajectory-scaled
`cached_sigma`	pre-built covariance matrix, reused across parallel workers
`makePositiveDefinite`	positive-definiteness repair switch
`n_cores`	worker count; `-1` auto-detects
`save_chunks`	progressive checkpointing for long runs

Sample size and randomization paths

Rule 4 fixes N as the total. The complication is that `modelparam$N` does not mean the same thing at every level of the call stack, and two families of driver read it differently.

The two levels. `generateData` generates one path: it takes a single `trialdesign` and treats `N` as the count for that path alone. `generateSimulatedResults` generates a whole design: it takes `N` as the total and allocates it, `Ns <- N %/% nP` with the remainder spread one per path. A driver that loops over `td$trialpaths` itself is therefore working at the lower level and must divide the total before the loop. A driver that passes the same full `N` to every iteration generates $N \times P$ participants, and its reported N is a per-path count wearing the wrong name.

Path counts. P is a property of the design, not a free parameter.

Design	P	Paths
OL	1	on drug throughout
CO	2	drug-then-placebo, placebo-then-drug
OL+BDC	2	discontinuation at either of two visits

Design	P	Paths
Hybrid	4	two discontinuation points crossed with two crossover orders

Paper 06 also uses a reduced single-path OL+BDC variant for its smaller cells and a two-path sixteen-visit variant for its larger ones, so P changes within that paper's sample-size axis.

Which drivers allocate and which duplicate. Established by inspection of every driver reachable from a current manuscript, on 2026-08-05.

Driver	Reads `N` as	Mechanism
`R/generateSimulatedResults.R`	total	`Ns <- N %/% nP` plus remainder
`analysis/scripts/carryover-sensitivity/simulation.R`	total	`generate_data_multi_path`, `floor(N / n_paths)`
`implementations/nof1power/R/simulation.R`	total	same integer split
`analysis/scripts/test-procedure-design-sensitivity/01-study1-test-procedure.R`	total	expression `N1`
`analysis/scripts/quick-sim/*.R`	per path	hand-rolled `for (g in seq_along(paths))` passing one `mp`
`analysis/scripts/component-decomposition/*.R`	per path	same shape
`analysis/scripts/gompertz-evaluation/02-faithful-trajectory-sweep.R`	total	full-trajectory

Status by paper. The manuscripts report totals throughout, and as of 2026-08-08 the drivers and stored output for papers 01, 06, 07 and 09 were brought into line with them. One entry is weaker than the others: the driver behind paper 01's Section 3.3 robustness run was not located, so its per-path reading is taken from that paper's own statement of its convention rather than from inspected code.

Paper	Driver was	Now
01	per path	driver allocates the total; re-run at a matched $N = 70$ for all three designs
02, 05, 08, 10, 11	total	already correct; untouched
03, 04	coincident (single-path or path-free DGP)	unaffected

Paper	Driver was	Now
06	per path	all drivers allocate the total; stored `N` migrated to totals; see the caveat below
07	per path	driver allocates the total; re-run; stored `N` added as 70
09	per path	driver allocates the total; re-run at a matched $N = 70$ for all four designs

How the corrections were validated. The work ran in two stages, and it is worth keeping them apart. The first stage corrected the label only. Re-expressing a grid in totals and allocating across paths leaves the per-path counts unchanged, so the generated data should be identical and nothing but the recorded `N` should move. That was checked before each re-run by generating one replicate both ways and comparing: for every driver converted, the drawn data and the fitted coefficients were identical. The re-runs then reproduced the published cell power values exactly, in all 36 cells of paper 01, all 16 of paper 07 and all 16 of paper 09, with per-replicate coefficients moving by at most 7×10^{-9} (paper 01), 0 (paper 07) and 4×10^{-5} (paper 09), which is `lme` convergence-tolerance noise on identical data.

The second stage, for papers 01 and 09 only, changed the sample sizes themselves to match the designs, so it was expected to change results and did. Its internal check is that the cells whose totals were already 70 must still reproduce exactly while the others move: in paper 01 the CO and OL+BDC cells came back bit-identical and only the Hybrid column changed, by up to 0.233 in power.

Paper 06 was migrated, not re-run. Study A alone holds 200,000 stored fits, so re-running it to change a column that the equivalence check proves is mislabeled would cost hours to reproduce the same numbers. Instead `analysis/scripts/component-decomposition/ migrate-n-to-total.R` recoded the stored `N` in place, 109 files across five output directories, with pre-migration copies in `analysis/.n-migration-backup/` and read-back verification of row counts and of the absence of legacy values. A column-by-column comparison against the backups confirms that nothing but `N` changed. The mapping is per directory because the studies differ in path count: the study-A and contamination grids switch to the two-path design above a threshold, so 100 and 150 became 200 and 300 while 35 and 70 were already single-path totals; study-B recovery is two-path throughout, so 70 and 150 became 140 and 300; study-B balanced is single-path and was left alone.

All four paper-06 drivers are now converted: `run-study-a-prod.R`,

`run-study-contam.R` and `run-study-b-recovery.R` read `N` as a total and allocate it, and `run-study-b-balanced.R` needed nothing because its balanced-placebo design has a single path. Each conversion was checked two ways. An equivalence gate regenerated one replicate per design type under both parameterizations and got identical coefficients, standard errors, p-values and formula-dropped status. Then every stored checkpoint was matched against the new grid's `N` for its `cell_id`: 18 of 18 for the contamination study and 8 of 8 for study-B recovery, no mismatches, so each driver agrees with the data already migrated on disk. Where a fit function takes an `N` argument it is passed the per-path count, which is what it carried before, so the formula ladders keyed on it are unaffected. Study-B recovery's three fits declare the argument but never read it; it is kept per-path there for consistency with the other drivers.

The substantive problem was fixed on 2026-08-08. Fixing the per-path count across designs with different P leaves a comparison that is not matched on N, so papers 01 and 09 were re-run with every design at a common total of $N = 70$. This changes numbers rather than labels, and it changed conclusions in both papers.

In paper 01 only the Hybrid column moved, since CO and OL+BDC were already at 70 and reproduced exactly. Hybrid power fell from about 0.99 to 0.84, which took it off the ceiling and roughly tripled its measured carryover loss: 1.4 to 6.9 percent under mean moderation and 5.1 to 15.4 percent under covariance moderation. The paper's headline architecture contrast survives, with the difference-of-differences z at Hybrid moving from 3.96 to 2.80 and OL+BDC unchanged at 7.64, but its claim that mean-moderation losses are uniformly 1 to 3 percent did not, and the Hybrid design is no longer saturated anywhere in the grid.

In paper 09 the OL and hybrid rows moved. The happy-accident ordering held, hybrid 0.968 above OL+BDC 0.950 above crossover 0.854 above OL 0.088 at every dropout pattern, but the hybrid margin over OL+BDC collapsed to within one MCSE, and the paper's second conclusion, that the most powerful design is also the most dropout-robust, was withdrawn: hybrid loss under 40 percent biased dropout rose from 0.4 to 5.8 percentage points, level with OL+BDC's 5.6, so the earlier near-immunity was a ceiling effect. The OL row also stopped coinciding with α, moving from 0.022-0.046 to 0.086-0.112, which showed that reading it as a null-calibration check had been a numerical coincidence at one sample size rather than a property of the design.

The lesson worth carrying: an unmatched design comparison does not merely inflate one design, it can manufacture a robustness finding out of a ceiling effect.

Data columns

`ptID` (participant), `Sx` (outcome, the Y_{it} above), `bm` (biomarker), `bm_z` (standardized biomarker), `Dbc` (continuous exposure indicator, $D_{bc,it}$), `tsd`, `tod`, `tpb`,

`t_wk` (time in weeks), `path` (randomization path).

Stored factor levels and their reporting labels

Column	Stored values	Reported as	Mapped?
`spec`	`E1`, `E2`, `E3`	E1, E2, E3	no, identical
`carryover_form`	`exponential`, `linear`, `weibull`	as stored	no, identical
`design`	`CO`, `Hybrid`, `OLBDC`	CO, Hybrid, OL+BDC	only `OLBDC` to `OL+BDC`
`dgp_arch`	`mean_moderation`, `mvn`, `combined`	Mean, Covariance, Dual	yes; these are package API values

The `spec` column stored `A1`, `A2`, `A3` until 2026-07-31. Twenty-one `.rds` files were migrated to `E1/E2/E3` and the producing drivers updated to emit them, so the display-time mapping that previously bridged the two has been retired. Pre-migration copies are in `analysis/.spec-migration-backup/`. Output archived before that date, including anything under a paper's own `archive/`, still uses the old values.

Summary-table columns

`n_reps`, `n_converged`, `converged_frac`, `non_convergence_rate`, `power`, `bias`, `mean_estimate`, `true_value`, `empirical_se`, `mse`, `coverage`, and an `mc_se_`-prefixed Monte Carlo standard error for each (`mc_se_power`, `mc_se_bias`, `mc_se_mse`, `mc_se_coverage`, `mc_se_non_convergence`).

Part 3. Controlled vocabulary and glossary

The project lexicon at `docs/29-nof1-precision-medicine-lexicon.md` holds the full terminology set (241 terms across 13 thematic sections, with cross-field synonyms). The entries below are the subset carrying a fixed, enforced meaning in this compendium.

Trial designs

OL (open-label). Unblinded titration; no off-drug contrast, so it supplies no within-subject interaction information. **BDC** (blinded discontinuation). A blinded switch to placebo after open-label response. **OL+BDC**. Open-label

titration followed by blinded discontinuation. **CO** (crossover). Traditional two-period within-subject crossover. **Hybrid**. The Hendrickson (2020) design: open-label titration, blinded discontinuation, then a brief crossover. The program's reference design. Capitalize it. **Aggregated N-of-1 trial**. Many single-patient on/off series pooled into one mixed model.

Define each abbreviation at first use in every paper. Each design's randomization-path count is given in Part 2, under sample size and randomization paths, and is not repeated here.

Data-generating architectures

Named by the channel that carries the interaction. Single letters are no longer used.

Name in prose	Legends and tables	Data value	Former label
the mean-moderation architecture	Mean	`mean_moderation`	Architecture A
the covariance-moderation architecture	Covariance	`mvn`	Architecture B
the dual-channel architecture	Dual	`combined`	Architecture C

Mean-moderation. The biomarker scales the treatment effect in the conditional mean; the interaction lives in the first moment. The near-universal convention in the trial-simulation literature. **Covariance-moderation.** The biomarker-response correlation is treatment-state dependent; the interaction lives in the second moment. The Hendrickson formulation, and the one whose power is most eroded by carryover. **Dual-channel.** Both channels active with independent weights $c_{bm,A}$ and $c_{bm,B}$; the other two are its single-channel limits. A data-generating construct only: the two weights are not separately identifiable from a fitted `bm:Dbc` coefficient.

Named data-generating processes (added 2026-10-08)

The current best DGP is called **refined** (user decision, 2026-10-08): "the refined process" in prose, "Refined" in legends and tables, against "the published process" / "Published" for Hendrickson's code.

Name in prose	Meaning	Defined in
the published process	Hendrickson's `generateData()` and `buildtrialdesign()` at commit 58b32a9: compound symmetry within components ($\rho_{cs} = 0.8$), constant cross-component correlation, step coupling, compounding carryover recursion, time on drug restarting at zero on re-exposure	Paper 01
the refined process	The recommended DGP: graded coupling (0 before first exposure), separable AR(1) with $\rho = 0.7$ and $c_1 = 0.2$, anchored carryover, a stop on matrices that are not positive definite, randomized path allocation; re-exposure as in the published process; optional prognostic couplings, phase-specific measurement error, Weibull decay and dropout	Paper 01, Table `tab:spec`
the published construct	The covariance construct of the published process alone (configuration A: compound symmetry within components, $\rho_{cs} = 0.8$, constant cross-component correlation), with whatever coupling the context states; not the whole published process (added 2026-10-09)	CM, glossary; used in CM, `docs/43`, `docs/56`

Name in prose	Meaning	Defined in
configuration A to F	The covariance and coupling constructs compared in `docs/36` (A published; B AR(1) with the `covar` cross block, step coupling; C separable AR(1) with graded coupling; D mean moderation; E all three; F AR(1) with the `covar` cross block, graded coupling)	`docs/36`, Section 6
`orig`, `covar`	Construct labels of the whitepapers. `orig`: the published process's construct as implemented in a given copy of the code (in `docs/54` the 2024 vendored copy, not commit 58b32a9, so the document states which); `covar`: the package's covariance-moderation construct, AR(1) within component with the decaying cross-component block (configuration F at the package's parameters) (added 2026-10-09)	`docs/32`, `docs/34` to `docs/38`, `docs/54`

Use **the refined process** for the DGP of the stress test and the recommendation. Use **configuration C** only for the covariance and coupling construct itself, as in the `docs/36` comparison, and state ρ with it (`docs/36` evaluates it at $\rho = 0.8$; the refined process uses $\rho = 0.7$). Configuration C is not the refined process: it does not by itself include anchored carryover, the positive-definite stop or the options. Do not confuse configuration C with the former label "Architecture C" (the dual-channel architecture).

Analysis specifications for carryover

E1. Binary on-drug indicator; carryover ignored. **E2.** Exposure-weighted continuous predictor D_{bc}, committing to an assumed decay form and half-life. **E3.** Binary indicator plus a lagged just-off-drug nuisance term; the classical Jones-Kenward crossover device, assuming no decay form.

The letter E stands for the exposure regressor that distinguishes them. These were labeled A1/A2/A3 until 2026-07-31; the stored data and the drivers were migrated to E1/E2/E3, so the labels now agree everywhere and nothing is mapped at display time.

Why not A/B/C and M1/M2/M3. The original scheme used A/B/C for architectures and A1/A2/A3 for specifications, colliding on the letter A within a single manuscript. Relabeling the specifications M1/M2/M3 removed that collision but created a second one: the calibration manuscript already uses M0-M3 for its working-covariance model ladder, and those labels are keys into archived data. E1/E2/E3 was free across the whole corpus when it was chosen, and naming the architectures by mechanism removes the class of collision entirely.

The extended E series (2026-10-09). The E labels are no longer free beyond E3: later documents extend the series, and every extension is listed here so that no label is reused with a second meaning.

Label	Meaning	Defined in
E4	binary drug state plus linear time since discontinuation (retired)	`docs/43` (from `implementations/original`)
E5	lag term crossed with the biomarker, L_{it} and $B_i L_{it}$ (retired)	`docs/43`, paper 02
E6	t_{sd} crossed with the biomarker (retired)	`docs/43`, paper 02
E7	E2 form with the half-life chosen by AIC	paper 02 (stored code), `docs/43`, CM
E9	paired-difference statistic: each participant's on-drug mean minus off-drug mean, Δ_i, regressed on the biomarker by OLS	paper 01, paper 02 (stored code), `docs/37`, CM
E9b	E9 with baseline counted as one more off-drug visit	paper 01, `docs/37`, CM

Label	Meaning	Defined in
E1cr2, E2cr2, E3cr2, E7cr2	the same formulas with the CR2 cluster-robust standard error	paper 02 (stored codes), CM

E8 is unassigned. E9 is a summary statistic rather than an exposure coding; it keeps its label because the label is a key in paper 02's stored data.

Response-curve families

Modified Gompertz (default), $y = m\,\exp(-d\,\exp(-r\,t))$ with a vertical offset so the curve passes through the origin; written $G(t_{od}; m, d, r)$ for a component's mean, with ceiling m (points), displacement d (none) and rate r (per week), and inflection time $t^* = \ln(d)/r$ (weeks). The code names are `maxr`, `disp` and `rate` (`respparam`); write the mathematical symbols in equations and the code names only in code (2026-10-09; the equation was previously written with the code names). A component subscript names the curve where needed (m_{BR}, G_{TV}). Alternatives: **logistic** (symmetric sigmoidal), **hyperbolic tangent** (fast-saturating), **piecewise-linear breakpoint** (no smoothness assumption).

Missing data and dropout

MCAR, **MAR**, **MNAR** in the standard Rubin sense. **Informative dropout.** Hazard depending on cumulative symptom worsening since baseline. **Happy accident.** The randomization-path selection effect by which informative dropout preferentially preserves the most informative crossover blocks.

Inference

LME. Linear mixed-effects model; the program's default analysis. **corCAR1.** Continuous-time AR(1) residual correlation structure. **GEE.** Generalized estimating equations; the marginal comparator. **Mancl-DeRouen**, **CR2.** Small-sample bias-corrected sandwich variance estimators. **RM-ANOVA.** Repeated-measures analysis of variance; the strict form requires a dichotomized biomarker. **Anti-conservative.** Realized Type I error above nominal. **Working covariance.** The assumed residual covariance of the LME; reserve **working correlation structure** for the GEE object.

Estimands and reporting

Biomarker-treatment interaction. The program's core estimand; equivalently the patient-by-treatment interaction, or heterogeneity of treatment effects (HTE) in the precision-medicine literature. **Predictive biomarker.** Predicts differential benefit, as opposed to a **prognostic** biomarker, which predicts outcome regardless of treatment. **ADEMP.** The Morris, White and Crowther (2019) reporting framework: aims, data-generating mechanisms, estimands, methods, performance measures. **MCSE.** Monte Carlo standard error; reported with every performance measure.

Enumeration conventions

Simulation studies. Number them (Study 1, Study 2, …) rather than lettering them. **Sensitivity sweeps.** Labels S1, S2, … are paper-local and are not comparable across papers. Cite them with the paper prefix when crossing manuscripts (05-S2, not S2).

Prose terminology

Symbol conventions are settled here; prose terminology is settled by `docs/30-terminology-consistency-plan.md`, executed 2026-07-30 with its D1 spelling decision locked to **US English**, overriding that document's original frequency-based recommendation of British forms. The two are complementary: this file governs mathematics and identifiers, that one governs surface forms and acronym expansion.

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